

Statistical Summary Report: Student Depression Dataset Analysis

1. Introduction

This report presents a comprehensive statistical analysis of the Student Depression Dataset, highlighting the central tendencies, distributions, and key insights from the data. This analysis is part of Task 9 (T9) in our project plan scheduled for April 12-14, 2025.

2. Dataset Overview

The dataset contains 27,901 records with 18 variables in the original dataset and 54 variables in the engineered dataset, including both categorical and numerical features focused on factors potentially associated with depression among students.

3. Descriptive Statistics for Numerical Variables

3.1 Demographics and Academic Variables

Variable	Count	Mean	Median	Std Dev	Min	Q1	Q3	Max
Age	27,901	25.82	25	4.91	18	21	30	59
CGPA	27,901	7.66	7.77	1.47	0	6.29	8.92	10
Work/Study Hours	27,901	7.16	8	3.71	0	4	10	12

3.2 Psychological and Stress Variables

Variable	Count	Mean	Median	Std Dev	Min	Q1	Q3	Max
Academic Pressure	27,901	3.14	3	1.38	0	2	4	5
Work Pressure	27,901	0.0004	0	0.044	0	0	0	5
Study Satisfaction	27,901	2.94	3	1.36	0	2	4	5
Job Satisfaction	27,901	0.0007	0	0.044	0	0	0	4
Financial Stress	27,901	3.14	3	1.44	1	2	4	5

3.3 Engineered Features

Variable	Count	Mean	Median	Std Dev	Min	Q1	Q3	Max
Overall_Stress	27,901	0.59	0.56	0.24	0	0.44	0.78	1
Academic_x_Satisfaction	27,901	9.04	8	5.85	0	4	12	25
JobSat_x_WorkPressure	27,901	0.0015	0	0.17	0	0	0	20

3.4 Target Variable

Variable	Count	Mean	Median	Std Dev	Min	Q1	Q3	Max
Depression	27,901	0.59	1	0.49	0	0	1	1

4. Distribution of Categorical Variables

4.1 Gender Distribution

- Male: 15,547 (55.7%)
- Female: 12,354 (44.3%)

4.2 Sleep Duration Distribution

- Less than 5 hours: 8,310 (29.8%)
- 5-6 hours: 6,183 (22.2%)
- 7-8 hours: 7,346 (26.3%)
- More than 8 hours: 6,044 (21.7%)
- Others: 18 (0.1%)

4.3 Dietary Habits Distribution

- Unhealthy: 10,317 (37.0%)
- Moderate: 9,921 (35.6%)
- Healthy: 7,651 (27.4%)
- Others: 12 (0.0%)

4.4 Suicidal Thoughts Distribution

- Yes: 17,656 (63.3%)
- No: 10,245 (36.7%)

4.5 Family History of Mental Illness

- No: 14,398 (51.6%)
- Yes: 13,503 (48.4%)

4.6 Top 10 Cities (by frequency)

1. Kalyan: 1,570 (5.6%)
2. Srinagar: 1,372 (4.9%)

3. Hyderabad: 1,340 (4.8%)
4. Vasai-Virar: 1,290 (4.6%)
5. Lucknow: 1,155 (4.1%)
6. Thane: 1,139 (4.1%)
7. Ludhiana: 1,111 (4.0%)
8. Agra: 1,094 (3.9%)
9. Surat: 1,078 (3.9%)
10. Kolkata: 1,066 (3.8%)

4.7 Top 10 Degree Programs (by frequency)

1. Class 12: 6,080 (21.8%)
2. B.Ed: 1,867 (6.7%)
3. B.Com: 1,506 (5.4%)
4. B.Arch: 1,478 (5.3%)
5. BCA: 1,433 (5.1%)
6. MSc: 1,190 (4.3%)
7. B.Tech: 1,152 (4.1%)
8. MCA: 1,044 (3.7%)
9. M.Tech: 1,022 (3.7%)
10. BHM: 925 (3.3%)

5. Depression Distribution and Correlations

5.1 Depression Score Distribution

- Not Depressed (0): 11,565 (41.5%)
- Depressed (1): 16,336 (58.5%)

5.2 Correlation with Depression

Variable	Correlation Coefficient
Academic Pressure	0.475
Financial Stress	0.364
Work/Study Hours	0.209
Study Satisfaction	-0.168
Age	-0.226
CGPA	0.022
Work Pressure	-0.003
Job Satisfaction	-0.003

6. Key Insights from Descriptive Statistics

- Depression Prevalence:** 58.5% of students in the dataset are classified as depressed, indicating a high prevalence in this sample.
- Age Distribution:** The average age is 25.8 years, with a standard deviation of 4.9 years, suggesting most participants are young adults.
- Academic Performance:** The average CGPA is 7.66 out of 10, with a relatively narrow distribution (standard deviation of 1.47).
- Stress Factors:** Academic pressure (mean of 3.14/5) and financial stress (mean of 3.14/5) show moderate to high stress levels among students.
- Sleep Patterns:** Nearly 30% of students report getting less than 5 hours of sleep, which could be a significant contributing factor to mental health issues.
- Suicidal Ideation:** A concerning 63.3% of students report having had suicidal thoughts, highlighting the severity of mental health challenges.
- Key Correlations:**
 - Academic pressure shows the strongest correlation with depression (0.475)
 - Financial stress is the second strongest correlate (0.364)
 - Age has a negative correlation (-0.226), suggesting younger students may be more vulnerable
 - Study satisfaction shows a negative correlation (-0.168), indicating it may be a protective factor

7. Next Steps for Analysis

Based on these descriptive statistics, the following areas merit deeper investigation:

- The relationship between academic pressure, financial stress, and depression
- Sleep duration and dietary habits as potential mediating factors

3. The high rate of suicidal thoughts and its relationship with depression scores
4. Age-related differences in depression vulnerability
5. Geographical variations in depression rates across different cities
6. Program of study (degree) and its relationship with depression scores

This statistical summary provides the foundation for more advanced analyses including correlation analyses, visualization dashboards, and eventually predictive modeling to be conducted in the subsequent tasks of the project.